

Assignment 1

Page No.	
Date	

Q.1 Write a note on significant digits.

- A digits use to express a number are called significant digits.
- The concept of significant digits has been introduced primarily to indicate the accuracy of a numerical value.

- Following statements describe in notion significant digits.

- 1] All non-zero digits are significant.
- 2] All zero occurring non-zero digits are significant figures.
- 3] Trailing zeros following a decimal point are significant digits.

Eg. 8.50, 65.8, 0.230

- 4] zeros before decimal point and preceding non-zero digits are not significant.

Eg 1) 0.0001234 (1234×10^{-7})
2) 0.001234 (1234×10^{-6})

Have 4 significant digits.

- 5] When the decimal point is not written trailing zeros are not considered to be significant.

E.g. 4500 (45×10^2) contains only 2 significant digits.

$$7.56 \times 10^4 = 3$$

$$7.560 \times 10^4 = 4$$

$$7.5600 \times 10^4 = 5$$

Q. 2. Write a note on Round off error:-

Round off errors occur when a fix no of digits are used to represent exact No. since the no are stored at any stage of computation. Round off error is introduced as they are of every arithmetic operation. Round the a number can be done in two ways.

a) chopping and other is known as symmetric Rounding.

1) Chopping-

In chopping extra digit are dropped this is called truncating the no. suppose we are using a computer with a fix word length of 4 digits. then a no then a no 412.7893 will be stored as 42.78 and they digit 93 will be dropped we can express the no 412.7893 in floating point form as.

$$\text{Eg} - x = 42.7893$$

$$\begin{aligned} &= 0.427893 \times 10^2 \\ &= (0.4278 + 0.93 \times 10^{-4}) \times 10^2 \\ &= 0.4278 \times 10^2 + 0.93 \times 10^{-2} \\ &= \text{fix} \times 10^2 + 93 \times 10^{-4} \end{aligned}$$

this can be expressed is general form as

$$\text{True } x = (\text{fix} + 93 \times 10^{-4}) \times 10^2$$

$$\text{True } x = \text{Approximate } x + \text{Error}$$

where,

$$\text{fix} = \text{the mantissa}$$

$$d \rightarrow \text{the length of mantissa permitted}$$

$$E \rightarrow \text{Exponent}$$

In chopping 93 is ignored intricately

$$\text{Error} = 93 \times 10^{-4} \text{ or } 93 \times 10^{-5}$$

$$\text{Error} = \text{True} - \text{Approximate}$$

ii) symmetric Round off error.

In the symmetric round off method the last significant digit is rounded the up by one if first discarded digit is larger or equal to 5 other wise the last written digit unchanged

eg. 412.7893 become $\Rightarrow 412.79$

$$412.7893 \Rightarrow 412.79$$

As before the value of rounded no can be expressed as

$$\begin{aligned} \text{True } x &= \text{fix} \times 10^2 + 93 \times 10^{-4} \\ \text{Approximate } x &= \text{fix} \times 10^2 \text{ and error} \\ &= 93 \times 10^{-4} \text{ or } 93 \times 10^{-5} \end{aligned}$$

- When $gx > 0.5$ the last digit in the mantissa is increased by one

$$\therefore \text{Approximate } x = (Fg + 10^d) \times 10^E$$

$$= Fg \times 10^E + 10^E \times 10^d$$

$$\therefore \text{Error} = \text{True } x - \text{Approximate } x$$

$$= Fg \times 10^E + g \times 10^{E-d} - Fg \times 10^E - 10^E \times 10^d$$

$$= (g \times 10^E - 10^E \times 10^d) + g \times 10^{E-d}$$

Q. 3 Write a note on Machine epsilon

Number since the round off introduced in a number when it is represented Floating point form is given by

chopping error $= g \times 10^{E-d} \cdot 0.5 < 1$
 where 'g' represents the truncated part of the number in normalised form, 'd' is the length of mantissa permitted and E is the exponent. The absolute error due to chopping is given by

$$\text{and } \epsilon_g = |g \times 10^{E-d}|$$

$$\text{Relative error} = \epsilon_z = \left| \frac{g \times 10^{E-d}}{F \times 10^E} \right|$$

- The relative error is maximum when "g" is maximum and E is minimum. we know that the maximum possible value of "g" is less than 1.0 and minimum possible value of F is 0.1.

- The absolute value of the relative error therefore satisfies.

$$\epsilon_r \leq \left| \frac{1.0 \times 10^E \times 10^d}{0.1 \times 10^E} \right| = 10^{-d+1}$$

The maximum relative error given above is known as machine epsilon

$\therefore$ Machine epsilon $= \epsilon = 10^{-d+1}$

similarly for a machine, which used symmetric round off

$$\epsilon_r \leq \left| \frac{0.5 \times 10^{E-d}}{0.1 \times 10^E} \right| = \frac{1}{2} \times 10^{-d+1}$$

and

$$\therefore \text{Machine epsilon} = \epsilon = \frac{1}{2} \times 10^{-d+1}$$

- The machine epsilon represent upper bound for the round off error due to floating point representation. data can be represented in the machine with d. significant decimal digits.

Q.4 Write a note on Truncation error.

⇒ Truncation error arises from using and approximate in place of mathematical error

- define - The errors which arises due to cutting of lengthy series at particular point

- These are the errors which are resulted due to representation of infinite quantity to find value for ex.

eg. $S = \sum_{i=1}^{\infty} a_i x^i$ is replaced by the

$$\text{Finite sum} = S = \sum_{i=1}^n a_i x^i$$

For example, consider,

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

- when we calculate and $\sin$ and angle of using its series, we cannot use all the terms for the complication we usually terminate the process after a certain term when the term is calculate.

- The terms truncated introduce an error which is called truncation error.

Q.5 Define Round-off and Round the following numbers to two decimal places

(i) 58.22421, (ii) 5.475, (iii) 5.485, (iv) 92.3742.

⇒

$$(i) \quad 58.22421 \Rightarrow 58.22$$

$$(ii) \quad 5.475 \Rightarrow 5.48$$

$$(iii) \quad 5.485 \Rightarrow 5.49$$

$$(iv) \quad 92.3742 \Rightarrow 92.37$$

Ex 6 IF $x = 3.4327$, Find the values of absolute and relative error if (i) x is truncated to 3 decimal places (ii) x is rounded off to 3 decimal places.

Soln:

Given = True value of $x = 3.4327$

(i) IF x is truncated to 3 decimal places.

$$x_a = 3.432$$

$$\begin{aligned} \text{absolute error} &= | \text{True } x - \text{Approximate } x | \\ &= | 3.4327 - 3.432 | \\ &= 0.0007 \end{aligned}$$

$$\text{Relative error} = \frac{|\text{True } x - \text{Approximate } x|}{\text{True } x}$$

$$= \frac{0.0007}{3.4327} \\ = 0.0002039$$

(ii) IF x is rounded off 3 decimal places

$$\text{True } x_t = 3.4327$$

$$\text{Approximate } x_a = 3.433$$

$$\text{absolute error} = |\text{True } x - \text{Approximate } x|$$

$$= |3.4327 - 3.433| \\ e_a = -0.0003$$

$$\text{Relative error} = \frac{|\text{True } x - \text{Approximate } x|}{\text{True } x}$$

$$= \frac{-0.0003}{3.4327} \\ e_r = -8.7394 \times 10^{-5} \\ = -0.00087394$$

Q. 7. A civil engineer has measured the height of a 10 Floor building as 2950 cm & the working height of each beam as 35 cm while the true values are 2945 cm and 30 cm respectively compare their absolute and relative errors.

= Solⁿ

True value of height of 10 Floor building is $x_t = 2945$

Approximate value of height of 10 Floor building is $x_a = 2950$

True value of beam = 30

Approximate value $x_a = 35$

$$\text{absolute error} = |x_t - x_a| \\ E_{a1} = 5 \text{ cm}$$

$$\text{Relative error} = \frac{5}{2945} \\ E_{r1} = 0.001697$$

$$x_{t2} = 30, x_{a2} = 35$$

$$E_{a2} = |x_t - x_a|$$

$$E_{a2} = 5 \text{ cm}$$

$$\text{Relative error} = \frac{|x_t - x_a|}{x_t} = \frac{5}{30} \\ = 0.001697 \\ = 0.1697 \times 10^{-2}$$

Q.8 Find the round off error in storing the number 752.6835 using a four digit mantissa.

→ solⁿ

$$752.6835 = 0.7526835 \times 10^3$$

$$\text{By four digit mantissa} = 0.7526 \times 10^3$$

Formula =

$$\begin{aligned} \text{True } x &= \text{Approximate } x \pm \text{Error} \\ &= Fx + 10^E + 9x \times 10^{E-d} \\ &= (Fx + 9x \times 10^{-d}) \times 10^E \end{aligned}$$

• chopping -

$$\begin{aligned} 0.7526835 \times 10^3 &= 0.7526 \times 10^3 + 0.0000835 \times 10^3 \\ &= 0.7526 \times 10^3 + 0.835 \times 10^{-1} \\ &= 0.7526 \times 10^3 + 0.835 \times 10^{-1} \end{aligned}$$

$$\text{True } x = 0.7526835 \times 10^3$$

$$\text{Approximate } x = 0.7526 \times 10^3$$

$$\text{Error} = 0.0835$$

• symmetric Rounding

$$9x > 0.5$$

$$\begin{aligned} \text{Approximate } x &= (Fx + 10^{-d}) \times 10^E \\ &= Fx \times 10^E + 10^{E-d} \\ &= 0.7526 \times 10^3 + 10^3 \times 10^{-1} \\ &= (0.7426 \times 10^{-4}) \times 10^3 \end{aligned}$$

F

$$\begin{aligned} \text{Error} &= (9x - 1) \times 10^{E-d} = (0.835 - 1) \times 10^{-1} \\ &= -0.165 \times 10^{-1} \\ &= \underline{\underline{0.0165}} \end{aligned}$$

Q.9 Find the truncation error in the result of the following Function for $x = \frac{1}{5}$ when we use

- First three terms
- First four terms
- First five terms

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!}$$

(a) Given :- upto 7 terms

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!}$$

$$\text{Truncation error of first three terms} = \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!}$$

$$= \frac{(0.2)^3}{6} + \frac{(0.2)^4}{24} + \frac{(0.2)^5}{120} + \frac{(0.2)^6}{720}$$

$$\begin{aligned} \text{Truncation error} &= 1.402 \times 10^{-3} \\ &= \underline{\underline{0.1402 \times 10^{-2}}} \end{aligned}$$

c] Truncation error when First Five terms are added

$$= \frac{(0.2)^5}{120} + \frac{(0.2)^6}{720}$$

$$= \underline{0.27566 \times 10^{-5}}$$

b] Truncation error when First Four terms added

$$= \frac{(0.2)^4}{24} + \frac{(0.2)^5}{120} + \frac{(0.2)^6}{720}$$

$$= \underline{0.694216 \times 10^{-4}}$$

Q.10 Find the relative error in $z = x - y$ when $x = 0.1234 \times 10^4$ & $y = 0.1232 \times 10^4$ as stored in a system with four digit mantissa.

Assignment 2

Page No.	
Date	

Ex1 Determine using bi-section method a root of the eqⁿ $\cos x - 1.3x = 0$ with accuracy 0.01
solⁿ

Given data:- $F(x) = \cos x - 1.3x = 0$ — (1)
Accuracy = 0.01

$$x_1 = \frac{a+b}{2}$$

$$\cos x - 1.3x = 0$$

let,

$$F(x) = \cos x - 1.3x$$

$$F(0) = 1 > 0$$

$$F(1) = -0.3001 < 0$$

locⁿ of root lies betⁿ

$$(0, 1)$$

$$F(1)$$

Iteration 1

By section method

$$x_0 = \frac{x_1 + x}{2} = \frac{0 + 1}{2} = 0.5$$

$$F(0.5) = 0.3499 > 0$$

$$\begin{array}{r} 0.3499 \\ - 0.3001 \\ \hline \end{array}$$

$$0.5$$

$$1$$

$$|1 - 0.5| = 0.5 > 0.01$$

Iteration	x_1	$F_1=f(x_1)$	x_2	$F_2=f(x_2)$	x_0	$F_0=f(x_0)$	Accuracy
1	0	1	1	-0.3001	0.5	0.3499	0.5
2	0.5	0.3499	1	-0.3001	0.75	0.0249	0.25
3	0.75	0.0249	1	-0.3001	0.875	-0.1376	0.125
4	0.75	0.0249	0.875	-0.1376	0.8125	-0.0563	0.0625
5	0.75	0.0249	0.8125	-0.0563	0.7812	-0.156	0.0312
6	0.75	0.0249	0.7812	-0.0156	0.7656	0.0046	0.0156
7	0.7656	0.0046	0.7812	-0.0156	0.7734	-0.0055	0.0078

Approximate root = 0.0078 < 0.01

Ex-2 Solve the eqn $xe^x - 3.1(\cos x) = 0$ by bisection method up to an accuracy of 0.01

Soln.

Given data.

$$f(x) = xe^x - 3.1(\cos x) = 0$$

Accuracy = 0.01.

Let,

$$f(x) = xe^x - 3.1(\cos x) \quad \text{--- (1)}$$

put, $x=0$ in eqn (1)

$$F(0) = -3.1 < 0$$

$$F(1) = 0.3812 < 0$$

$$F(2) = 11.68 > 0$$

$\therefore$ The location of root lies betn (1/2)

$$F(1) = 0.3812$$

$$F(0) = -3.1$$

$$F(2) = 11.68$$

Iteration

$$x_0 = \frac{x_1 + x_2}{2} = \frac{1 + 2}{2} = 1.5$$

$$F(1.5) = 6.2221 > 0$$

Iteration	x_1	$F_1=f(x_1)$	x_2	$F_2=f(x_2)$	x_0	$F_0=f(x_0)$	Accuracy $ x_1 - x_0 /x_1$
1	1	-0.3812	2	11.6800	1.5	6.2212	0.5
2	1	-0.3812	1.5	6.2212	1.25	1.2636	0.25
3	1	-0.3812	1.25	1.2636	1.125	0.3658	0.125
4	1	-0.3812	1.125	0.365	1.0625	-0.025	0.0625
5	1.0625	-0.025	1.125	0.365	1.0937	0.1055	0.0313
6	1.0625	-0.025	1.0937	0.655	1.0781	0.0691	0.0125
7	1.0625	-0.025	1.0781	0.0691	1.0703	0.0218	0.007
8	1.0625	-0.025	1.0703	0.218	1.0664	0.0167	

Approximate Root = 1.0664

Ex3 Using three iteration of bi-section method determine root of the eqn Initial guesses are

$$x_1 = 2.8 \text{ and } x_2 = 3$$

$$F(x) = -0.9x^2 + 1.7x + 2.5$$

Soln

$$\text{Given data: } f(x) = -0.9x^2 + 1.7x + 2.5 \quad \text{--- (1)}$$

Root lies betn

$$x_1 = 2.8, \quad x_2 = 3$$

Step 1: Check whether initial guesses are right or wrong

$$\text{For } x_1 = 2.8$$

$$\therefore F(x_1) = -(0.9)x(2.8)^2 + 1.7(2.8) + 2.5 \\ = \underline{0.204}$$

$$\text{For } x_2 = 3$$

$$F(x_2) = 0.9(3)^2 + 1.7(3) + 2.5 \\ = \underline{-0.5}$$

Iteration 1

$$x_0 = \frac{x_1 + x_2}{2} = \frac{2.8 + 3}{2} = \underline{2.9}$$

Iteration	x_1	$F(x_1)$	x_2	$F(x_2)$	x_0	$F(x_0)$
1	2.8	0.204	3	-0.5	2.9	-0.139
2	2.8	0.204	2.9	-0.139	2.85	0.0345
3	2.85	0.0345	2.9	-0.319	2.875	0.051562

$\therefore$ The root after 3rd iteration
 $x_0 = \underline{2.875}$

Ex4. (a) Find a root of the eqn $x^3 - 4x - 9 = 0$ using bisection method correct to three decimal places

(b) using bisection method, Find the negative root of the eqn $x^3 - 4x + 9 = 0$.
Soln.

(a) let,

$$F(x) = x^3 - 4x - 9$$

Since $F(2)$ is -ve and $F(3)$ is +ve, a root lies betn 2 and 3

$\therefore$ First approximate to the root is

$$x_1 = \frac{1}{2}(2+3) = 2.5$$

$$\text{Thus, } F(x_1) = (2.5)^3 - 4(2.5) - 9 = -3.375 \text{ i.e. -ve}$$

$\therefore$ The root lies betn x_1 and 3 Thus the second approximate to the root is

$$x_2 = \frac{1}{2}(x_1 + 3) = 2.75$$

$$\text{Then } F(x_2) = (2.75)^3 - 4$$

$\therefore$ The root lies betn x_1 and x_2 . Thus the third approximation to the root is

$$x_3 = \frac{1}{2}(x_1 + x_2) = 2.625$$

$$\text{Then } F(x_3) = (2.625)^3 - 4(2.625) - 9 \\ = 1.4121 \text{ i.e. +ve.}$$

The root lies betⁿ x_2 and x_3 . Thus the fourth approximate to the root is

$$x_4 = \frac{1}{2} (x_2 + x_3) = 2.6875$$

Repeating this process, the successive approximations are

$$\begin{aligned} x_5 &= 2.71875, \quad x_6 = 2.70313, \quad x_7 = 2.71094 \\ x_8 &= 2.70703, \quad x_9 = 2.70508, \quad x_{10} = 2.70605 \\ x_{11} &= 2.70654, \quad x_{12} = 2.70642. \end{aligned}$$

Hence the root is 2.7064

(b) If α, β, γ are the roots of given eqⁿ. then $-\alpha, -\beta, -\gamma$ are the roots of $(-x)^3 - 4(-x) + 9 = 0$

$\therefore$ The negative root of the given eqⁿ is the positive root of $x^3 - 4x + 9 = 0$ which we have found above to be 2.7064. Hence the negative root of the given eqⁿ is -2.7064

EX.5 Using the bisection method, Find an approximate root of the eqⁿ $\sin x = 1/x$ that lies betⁿ $x=1$ and $x=1.5$ (measure in radians) carry out computations upto the 7th stage.

Let $f(x) = x \sin x - 1$
We know that $1^\circ = 57.3^\circ$

$$\begin{aligned} \text{since } f(1) &= 1 \times \sin(1) - 1 = \sin(57.3^\circ) - 1 \\ &= -0.15849 \end{aligned}$$

$$\begin{aligned} \& f(1.5) &= 1.5 \times \sin(1.5) - 1 \\ &= 1.5 \times \sin(85.95^\circ) - 1 \\ &= 0.49625 \end{aligned}$$

a root lies betⁿ 1 and 1.5

$\therefore$ First approximate to the root is

$$x_1 = \frac{1}{2} (1 + 1.5) = 1.25$$

$$\begin{aligned} \text{Then } f(x_1) &= (1.25) \sin(1.25) - 1 \\ &= 1.25 \sin(71.625^\circ) - 1 \\ &= 0.8627 \text{ and } f(1) < 0. \end{aligned}$$

$\therefore$ A root lies betⁿ 1 and $x_2 = 1.125$

Thus the third approximate to the root is $x_3 = \frac{1}{2} (1 + 1.25) = 1.125$

$$\begin{aligned} \text{Then } f(x_2) &= 1.125 \sin(1.125) - 1 \\ &= 1.125 \sin(64.46^\circ) - 1 \\ &= 0.01509 \text{ and } f(1) < 0 \end{aligned}$$

$\therefore$ A root lies betⁿ 1 and $x_2 = 1.125$

Thus the third approximate to the root is $x_3 = \frac{1}{2} (1 + 1.125) = 1.0625$

$$\begin{aligned} \text{Then } f(x_3) &= 1.0625 \sin(1.0625) - 1 \\ &= 1.0625 \sin(60.88^\circ) - 1 \\ &= -0.0718 < 0 \text{ and } f(x_2) > 0 \end{aligned}$$

i.e. now the root lies between $x_3 = 1.0625$ and $x_2 = 1.125$

∴ Fourth approximation to the root is

$$x_4 = \frac{1}{2}(1.0625 + 1.125) = 1.09375$$

$$f(x_4) = -0.02836 < 0 \text{ and } f(x_2) > 0$$

The root betⁿ $x_4 = 1.09375$ and $x_2 = 1.125$

∴ Fifth approximation to the root is $x_5 = \frac{1}{2}(1.09375 + 1.125) = 1.109375$

$$\text{Then, } f(x_5) = -0.00664 < 0 \text{ and } f(x_2) > 0$$

∴ The root lies betⁿ $x_5 = 1.109375$ and $x_2 = 1.125$

Thus the sixth approximation to the root is

$$x_6 = \frac{1}{2}(1.109375 + 1.125) = 1.11719$$

$$\text{Then } f(x_6) = 0.00421 > 0 \text{ But } f(x_5) < 0$$

∴ The root lies betⁿ $x_5 = 1.109375$ and $x_6 = 1.11719$

Thus the seventh approximation to the root is $x_7 = \frac{1}{2}(1.109375 + 1.11719) = 1.11328$

Hence the desired approximation to the root is 1.11328

Ex 6 Find the root of the eqⁿ $\cos x = xe^x$ using the bisection method correct to four decimal places.

Solⁿ,

$$\text{let } f(x) = \cos x - xe^x$$

since $f(0) = 1$ and $f(1) = -2.18$, so a root lies betⁿ 0 and 1

∴ First approximation to the root is

$$x_1 = \frac{1}{2}(0+1) = 0.5$$

$$\text{Now } f(x_1) = 0.05 \text{ and } f(1) = -2.18$$

∴ the root lies betⁿ 1 and $x_1 = 0.5$

∴ second approximation to the root is

$$x_2 = \frac{1}{2}(0.5+1) = 0.75$$

$$\text{Now } f(x_2) = -0.86 \text{ and } f(0.5) = 0.05$$

therefore the root lies betⁿ 0.5 and 0.75

∴ Third approximation to the root is

$$x_3 = \frac{1}{2}(0.5+0.75) = 0.625$$

Now $f(x_3) = 0.96$ and $f(0.5) = 0.05$, therefore the root lies betⁿ 0.5 and 0.625

∴ Fourth approximation to the root is

$$x_4 = \frac{1}{2}(0.5+0.625) = 0.5625$$

Now $f(x_4) = -0.14$ and $0.5 = 0.05$, therefore the root lies betⁿ 0.5 and 0.5625

∴ fifth approximation is $x_5 = \frac{1}{2}(0.5 + 0.5025)$
 $= 0.5312$

Now $f(x_5) = -0.04$ and $f(0.5) = 0.05$, therefore
the root lies between 0.5 and 0.5312

∴ sixth approximation is $x_6 = \frac{1}{2}(0.5 + 0.5312)$
 $= 0.5156$

Hence the desired approximation to the
root is 0.5156

Ex-7 Find a positive real of $x \log_{10} x = 1.2$
using the bisection method.

Solⁿ

Let $f(x) = x \log_{10} x - 1.2$

Since $f(2) = -0.598$ and

$f(3) = 0.281$, so a root lies betⁿ
2 & 3

∴ First approximation to the root is

$$x_1 = \frac{1}{2}(2+3) = 2.5$$

Now $f(2.5) = -0.205$ and

$f(3) = 0.281$, therefore a root lies
betⁿ 2.5 and 3

∴ second approximation to the root is

$$x_2 = \frac{1}{2}(2.5 + 3) = 2.75$$

Now $f(2.75) = 0.011$ and $f(2.5) = -0.205$

therefore a root lies betⁿ 2.5 and 2.75

∴ Third approximation to the root

$$x_3 = \frac{1}{2}(2.5 + 2.75) = 2.625$$

Now $f(2.625) = -0.1$ and $f(2.75) = 0.008$

therefore a root lies betⁿ 2.625 & 2.75

∴ Fourth approximation to the root is

$$x_4 = \frac{1}{2}(2.625 + 2.75) = 2.687$$

Hence the desired root is 2.687

Ex-8 Find the positive root, betⁿ 0 & 1, of
the eqⁿ $x = e^{-x}$ to a tolerance of 0.05%.

Solⁿ

Let $f(x) = x e^x - 1 = 0$

we have, $f(0) = -1$ &

$$f(1) = e - 1$$

which is positive, hence a root exists
betⁿ 0 and 1, and

$$x_1 = \frac{0+1}{2} = 0.5$$

Because, $f(0.5) = -0.1755$, the root lies betⁿ
0.5 and 1.0

$$\text{Then, } x_2 = \frac{0.5 + 1.0}{2} = 0.75$$

Now, the tolerance % is given by

$$\% \text{ error} = \left| \frac{x_2 - x_1}{x_2} \right| \times 100$$

$$= \frac{0.25}{0.75} \times 100 = 33.33\%$$

since $F(x_2) = 0.5878$, the root lies betⁿ
0.5 and 0.75

∴

$$x_3 = \frac{0.5 + 0.75}{2} = \underline{0.625}$$

also,

$$E_2 = \left| \frac{0.625 - 0.75}{0.625} \right| \times 100 = 20\%$$